

Fig. 1: An efficient night lamp circuit

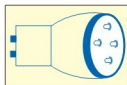


Fig. 2: LEDs arrangement in waste CFL

the serviceable electronic circuit fitted in the base of an 11-watt CFL.

For constructing this night lamp, remove the CFL glass tube and replace it with four white LEDs as described be-

low. You should be careful not to break the tube as it contains hazardous materials such as mercury. Carefully open the base of CFL holder using an appropriate tool. You can see the electronic ballast circuit on a circular PCB.

Use the components from the ballast circuit and a series combination

of four bright white LEDs as shown in Fig. 1. Remove all other components from the original ballast circuit. As per the requirement of light intensity in your room, you can increase the number of white LEDs up to eight.

As shown in Fig. 1, the full-wave bridge rectifier comprising diodes D1 through D4 converts AC voltage into DC voltage. Snubber capacitor C1 at the input reduces the line input voltage of 230V to a very low-level AC voltage. Series current-limiting resistor R2 and series inductor coil L1 avoid voltage spikes.

Fit the circular PCB in the waste CFL holder and your night lamp is ready for use. ●

Smart Exhaust Fan Switch

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Exhaust fans in washrooms can be connected to the lighting circuit so that these switch on and switch off with the electric light. But as washrooms are sometimes used only for washing the face or hands, it is not always necessary to switch on the exhaust fan with the light. Advanced techniques such as humidity sensor switching and PIR motion sensor switching can also be used to control

exhaust fans but these are not affordable for everyone.

The simple exhaust fan circuit presented here automatically determines whether there is really a need to start the fan and reacts appropriately. The circuit is powered from an AC mains adaptor that is capable of delivering an output supply of 6V DC.

The heart of the circuit (Fig. 1) is an ordinary reed switch (S1) that is mounted on the cistern. The reed switch is controlled by a small magnet

fitted on the toilet lid. When the lid is upright, the reed switch is closed by the magnet. One input of gate N1 of IC1 swings high. The resulting pulse at the output of gate N2 is inverted by gate N3, triggering the timer 555 (IC2). As a result, the relay (RL1) energises, causing the exhaust fan to run for a time period determined by the combination of VR1 and capacitor C3. After this time is over, the exhaust fan automatically switches off.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Mount the 5V relay (refer Fig. 2) on the PCB and connect relay contacts to the exhaust fan through external wires. Fig. 3 shows the proposed fitting of the magnet on the lid and the reed switch on the cistern. ●

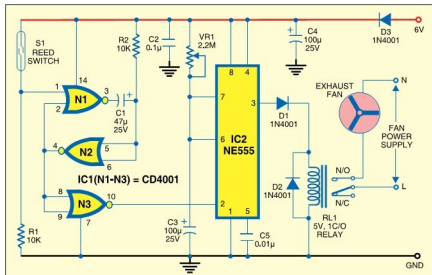


Fig. 1: Circuit of exhaust fan switch



Fig. 2: 5V DC 1C/O relay



Fig. 3: Proposed fittings